Humanoid Walking Stabilization via Full Stepping Strategy based on Three-Dimensional Divergent Component of Motion

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***Abstract*— Balance control has been widely studied to achieve the stable operation of humanoid robots. Among the various strategies, step time adjustment remains challenging, as previous methods rely on the dynamics of the linear inverted pendulum model, which limits applicability. In this paper, a novel method that enables the full stepping strategy even during vertical motion is proposed. By enabling contact schedule modifications within a single model predictive control framework, a Mixed Integer Programming (MIP) based step location and time adjustment is achieved. To alleviate the computational burden, problem-specific simplification replaces the tree-search procedure in the general MIP solution process. Throughout the proposed method, the humanoid can withstand disturbances, even during the operation that includes vertical motion. The proposed method is validated through simulations and real robot experiments in walking and loco-manipulation scenarios.**

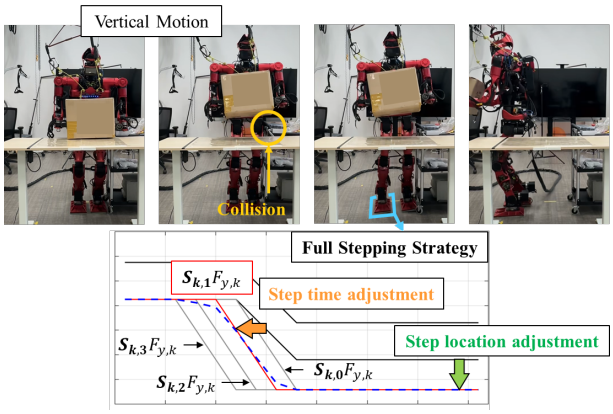


Fig. 1. Graphical Abstracts of Humanoid Walking Stabilization through the proposed full stepping strategy.

I. INTRODUCTION

Humanoid robots are expected to perform various tasks in human-centered environments, owing to their morphological similarity to humans. With the recent growth in interest, the complexity of assigned tasks and the diversity of operating environments for humanoid robots continue to increase. Under these circumstances, bipedal locomotion should support manipulation with accuracy and stability. These entail precise tracking of planned footsteps for task execution and collision-free navigation on diverse terrains, as well as maintaining robust balance against unexpected and unperceived disturbances to ensure safe operation.

In recent years, Reinforcement Learning (RL) has attracted significant attention for humanoid locomotion. In RL, balancing behaviors are relatively easy to obtain, induced by the perturbations introduced during training [1]. On the other hand, in footstep tracking, RL-based approaches have often shown limited performance despite the use of explicitly designed rewards. To mitigate this issue, model-based knowledge was incorporated into an RL-based controller by generating footsteps based on the Linear Inverted Pendulum Model (LIPM), and further, by introducing rewards that promote LIPM-like behavior [2]. Although the proposed approach achieved improved footstep tracking performance, reliance on LIPM limits its applicability to a narrow range. Nonetheless, the findings indicate that exploring alternative dynamic models may enable meaningful extensions in RL-based methods.

Model-based approaches have been widely studied over the past decades, drawing on dynamics and employing optimization techniques. Although model-based methods achieve high footstep-tracking performance, the complexity of dynamics and limited computational capacity make robust balance control challenging. To mitigate the difficulties, model-based strategies relied on reduced-order models.

LIPM is considered one of the most easily applicable models among them. Numerous studies proposed balance controllers utilizing the Divergent Component of Motion (DCM), the unstable part of LIPM dynamics. Those controllers compute the Zero Moment Point (ZMP), which is another representation of the Ground Reaction Force (GRF), along with step location and time adjustments to ensure the feasibility of the GRF [3]–[5]. However, reliance on LIPM constrains vertical motion, thereby limiting its applicability and hindering adaptability to diverse tasks and environments.

To enhance applicability and adaptability, the Variable-Height Inverted Pendulum Model (VHIPM) was introduced [6], [7]. Several studies based on VHIPM were proposed to stabilize the bipedal walking, but the control inputs were constrained to the use of GRF and step location adjustment [8], [9]. Furthermore, the inherent nonlinearity of VHIPM dynamics poses challenges for implementation.

Meanwhile, by analyzing the dynamics of VHIPM [10], [11], the Floating-base Inverted Pendulum Model (FIPM)

This work was supported by the Ministry of Trade, Industry and Resources (MOTIR) and the Korea Institute for Advancement of Technology (KIAT) through the Industrial Innovation Infrastructure Program (No. RS-2025-25460896) and by the Basic Science Research Program through the NRF, funded by the Ministry of Education (RS-2023-00274280).

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was derived. The nonlinearity in VHIPM dynamics is transferred to the equality constraints of FIPM, yielding linear dynamics with nonlinear constraints. The three-dimensional Divergent Component of Motion (3D-DCM) [12], [13], a 3D expansion of DCM, can also be derived from linear FIPM dynamics. Balancing methods based on FIPM and 3D-DCM were formulated [14], [15], but their functionality was also limited to GRF and step location adjustment, leaving step time adjustment unresolved. Although step time adjustment is essential for disturbance rejection in tasks involving vertical motion, its realization has remained largely underdeveloped. Integral-based methods that enable step time adjustment in LIPM under strict assumptions [3], [5] struggle to be extended due to the nonlinearity of the FIPM constraint.

Inspired by the step location adjustment variable in [5], [15]–[18], and the Mixed Integer Programming (MIP) formulations presented in [19], [20], this paper proposes a novel method that enables a real-time full stepping strategy, while involving vertical motion of the humanoid robot. To the best of our knowledge, this is the first method that achieves full stepping capability, step location and time adjustments, even during vertical motion. The contributions are as follows: 1) An MPC framework based on FIPM to control 3D-DCM with a full stepping strategy is proposed. The Virtual Repellent Point (VRP), step location, and time adjustment are calculated considering the underlying dynamics among them. 2) To enhance the real-time feasibility, the problem-specific simplification during the decision process of MIP is introduced, instead of utilizing a tree search-based general solution strategy. Furthermore, the number of variables for step time adjustment is reduced by considering the feasible minimum step time, in contrast to previous methods that parameterize all control ticks within the prediction horizon. 3) The effectiveness of the proposed method is validated through simulations and real robot experiments, including the loco-manipulation scenario that involves vertical motion to carry the object. The simulations and experiments showed that the proposed method improved the balancing capacity even during vertical motion.

II. Fundamentals

A. Floating-base Inverted Pendulum

Using Newton's equations of motion, the dynamics between the Center of Mass (CoM) and the ZMP can be derived. For simplicity, the centroidal angular momentum is not considered.

$$\ddot{\mathbf{c}} = \lambda(\mathbf{c} - \mathbf{p}_{zmp}) + \mathbf{g}, \tag{1}$$

where $\mathbf{c} = [c_x\, c_y\, c_z]^T, \mathbf{p}_{zmp} = [p_{zmp,x}\, p_{zmp,y}\, p_{zmp,z}]^T, \lambda$ and $\mathbf{g} = [0\, 0\, -g]^T$ denote the CoM position, ZMP position, normalized stiffness, and gravity vector, respectively. (1) should hold the following constraints for GRF.

$$\mathbf{p}_{zmp}^{min} \leq \mathbf{p}_{zmp} \leq \mathbf{p}_{zmp}^{max}, \tag{2}$$

$$\lambda \geq 0. \tag{3}$$

The ZMP should lie within the Supporting Polygon (SP), and the normalized stiffness should be positive if the robot is assumed not to lose contact with the ground. The equation for normalized stiffness and the CoM clearly shows it.

$$\lambda = \frac{\ddot{c}_z + g}{c_z}. \tag{4}$$

With contact assumption, acceleration of CoM in z direction should be bigger than $-g$, and normalized stiffness λ should be positive. As can be seen in (4), (1) is nonlinear since normalized stiffness λ is time-variant. The nonlinear dynamics in (1) is referred to as VHIPM dynamics, which should hold linear constraints in (2), (3).

To linearize (1), the Enhanced Centroidal Momentum Pivot point (eCMP) and VRP concepts are utilized [10], [11]. By replacing ZMP with eCMP, (1) is linearized as follows:

$$\mathbf{p}_{ecmp} = \lambda b^2 \mathbf{p}_{zmp} + (1 - \lambda b^2)\mathbf{c}, \tag{5}$$

$$\ddot{\mathbf{c}} = \frac{1}{b^2}(\mathbf{c} - \mathbf{p}_{ecmp}) + \mathbf{g}, \tag{6}$$

where $\mathbf{p}_{ecmp} = [p_{ecmp,x}\, p_{ecmp,y}\, p_{ecmp,z}]^T$ and b denote the eCMP position and time-constant of 3D-DCM dynamics, respectively. eCMP is the point located on the straight line starting from CoM and heading to ZMP, which encodes the effect of all external forces. The time-constant of 3D-DCM dynamics, b, is calculated as $\sqrt{c_{z,0}/g}$, where $c_{z,0}$ is the initial CoM height.

To simplify (6) further, eCMP is replaced by VRP as:

$$\mathbf{p}_{vrp} = \mathbf{p}_{ecmp} - b^2\mathbf{g}, \tag{7}$$

$$\ddot{\mathbf{c}} = \frac{1}{b^2}(\mathbf{c} - \mathbf{p}_{vrp}), \tag{8}$$

where $\mathbf{p}_{vrp} = [p_{vrp,x}\, p_{vrp,y}\, p_{vrp,z}]^T$ denotes the VRP position. VRP is the point located above eCMP at constant height, which encodes the effect of all external forces and gravity. Combining (5) and (7), the relation between ZMP and VRP is derived as:

$$\mathbf{p}_{vrp} = \lambda b^2 \mathbf{p}_{zmp} + (1 - \lambda b^2)\mathbf{c} - b^2\mathbf{g}. \tag{9}$$

Meanwhile, even after linearization, it still should hold the constraints in (2), (3). To utilize VRP as a control input while holding the constraints, (9) is additionally considered as an equality constraint. The linearized dynamics in (8) is referred to as FIPM dynamics, and the FIPM constraints correspond to (2), (3), and (9). Based on FIPM (illustrated in Fig. 2), the 3D-CoM trajectory planner in III-A and the 3D-DCM dynamics in II-B are derived.

B. Three Dimensional Divergent Component of Motion

After the introduction of 3D-DCM in [12], [13], Caron et al. investigated the relation between CoM dynamics and 3D-DCM, and clarified the distinction from the time-varying DCM, also known as 4D-DCM [11], [21]. This paper focuses on the utilization and derivation of 3D-DCM, using FIPM dynamics and decomposition for DCM [22].

FIPM dynamics (8) can be rewritten into a first-order

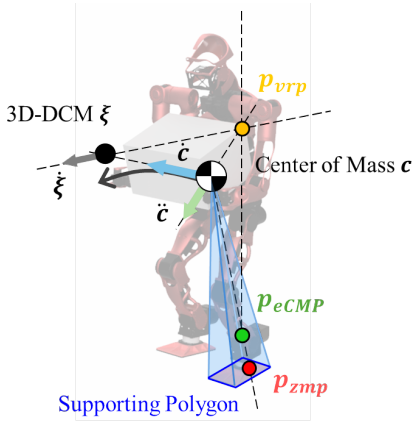


Fig. 2. Dynamics and Constraints of FIPM. The relation between ZMP, eCMP, VRP, CoM, 3D-DCM, and SP is illustrated.

linear system utilizing VRP as a control input.

$$\begin{pmatrix}\dot{\mathbf{c}}\\ \ddot{\mathbf{c}}\end{pmatrix} = \begin{bmatrix}\mathbf{0} & \mathbf{I}\\ \frac{1}{b^2}\mathbf{I} & \mathbf{0}\end{bmatrix}\begin{pmatrix}\mathbf{c}\\ \dot{\mathbf{c}}\end{pmatrix} + \begin{bmatrix}\mathbf{0}\\ -\frac{1}{b^2}\mathbf{I}\end{bmatrix}\mathbf{p}_{vrp}. \tag{10}$$

$\mathbf{I}$ and $\mathbf{0}$ denote 3×3 identity and zero matrices. Applying the following change of coordinate $\mathbf{S}$ to (10), FIPM dynamics is split into two parts, one stable and one unstable,

$$\begin{pmatrix}\zeta\\ \xi\end{pmatrix} = \mathbf{S}^{-1}\begin{pmatrix}\mathbf{c}\\ \dot{\mathbf{c}}\end{pmatrix}, \mathbf{S}^{-1} = \begin{pmatrix}\mathbf{I} & -b\mathbf{I}\\ \mathbf{I} & +b\mathbf{I}\end{pmatrix}. \tag{11}$$

The unstable part $\boldsymbol{\xi}$ is referred to as 3D-DCM. 3D-DCM dynamics with CoM and VRP is derived as follows:

$$\boldsymbol{\xi} = \mathbf{c} + b\dot{\mathbf{c}}, \tag{12}$$

$$\dot{\boldsymbol{\xi}} = \frac{1}{b}(\boldsymbol{\xi} - \mathbf{p}_{vrp}), \tag{13}$$

where $\boldsymbol{\xi} = [\xi_x\ \xi_y\ \xi_z]^T$ denote the position of 3D-DCM.

Based on dynamics and constraints for 3D-DCM, an MPC framework in Section III-B is derived. See [10]–[13] for more details about FIPM and 3D-DCM.

III. Walking Control Framework

Fig. 3 shows the schematic diagram of overall framework. Main goal of framework is to control 3D-DCM, $\boldsymbol{\xi}_k$. At first, the foot step planner calculates the reference footstep positions, $\mathbf{F}^{ref} = [\mathbf{F}_x^{ref}\ \mathbf{F}_y^{ref}\ \mathbf{F}_z^{ref}]^T$, with input target velocity and nominal step duration, T_{step}^{norm} or with terrain information. Second, the reference footstep positions, $\mathbf{F}^{ref}$, the adjusted step position, $\mathbf{F}_k = [\boldsymbol{F}_{x,k}\ \boldsymbol{F}_{y,k}\ \mathbf{0}]^T$, and the adjusted step time, T_{step}^{opt}, are delivered to the 3D-CoM trajectory planner to generate the 3D-CoM trajectory for MPC horizon N_h. Utilizing (12) and 3D-CoM trajectory, the reference 3D-DCM trajectory, $\boldsymbol{\Xi}_{k+1}^{ref} = [\boldsymbol{\xi}_{k+1}^{ref}\ \boldsymbol{\xi}_{k+2}^{ref}\cdots\boldsymbol{\xi}_{k+N_h}^{ref}]^T \in \mathbb{R}^{3N_h}$ is calculated. Next, based on the reference 3D-DCM trajectory and measured current 3D-DCM, $\boldsymbol{\xi}_k$, 3D-DCM MPC optimizes the control inputs, the derivative of VRP, $\dot{\mathbf{P}}_{vrp,k} = [\dot{\mathbf{p}}_{vrp,k}\ \dot{\mathbf{p}}_{vrp,k+1}\ \cdots\ \dot{\mathbf{p}}_{vrp,k+N_h-1}] \in \mathbb{R}^{3N_h}$, and step adjustment variables, $\boldsymbol{f}_k = [\boldsymbol{f}_{x,k}\boldsymbol{f}_{y,k}\ 0] \in \mathbb{R}^{3m}$. m is the number of predefined range of allowable step time adjustment. Step location $\mathbf{F}_k$ and step time T_{step}^{opt} are calculated using step adjustment variables, and sent to foot trajectory planner to calculate reference foot position/orientation, $\mathbf{e}_k^{des}$, and to 3D-CoM trajectory planner. Calculated VRP, $\mathbf{p}_{vrp,k}$, desired foot trajectory, $\mathbf{e}_k^{des}$, and desired CoM trajectory, $\mathbf{c}_k^{des}$ are sent to the joint control module to calculate desired lower body joint angle, $\mathbf{q}_{lb,k}^{des}$, and to calculate the joint torque command, $\boldsymbol{\tau}^{cmd}$, which is the sum of feed forward torque, calculated in whole-body controller [23], and PD feedback torque.

A. 3D-CoM Trajectory Planner based on FIPM

In this work, an MPC framework [24] based on FIPM dynamics with eCMP (6) is employed as a 3D-CoM trajectory planner. To ensure stability of the generated trajectory, the derivative of eCMP is used as a control input, and intrinsic stability constraints are incorporated [25], [26]. The primary difference from [24] lies in the treatment of the eCMP constraint. As shown in [24] and Fig. 2, the feasible eCMP region is represented as a polyhedral cone defined by lines connecting the CoM to the vertices of SP. While [24] approximates this region with a maximum inscribed box, we formulate the polyhedral constraint directly as quadratic constraints, resulting in a Quadratically Constrained Quadratic Programming (QCQP) that fully exploits the admissible eCMP region. As the derivative of VRP coincides with that of eCMP, (7), it is used as a control input for a detailed derivation in the APPENDIX.

B. 3D-DCM MPC with Full Stepping Strategy

In this section, one of the main contributions of this paper, the 3D-DCM MPC with full stepping strategy, is introduced. The goal of MPC is to track the reference 3D-DCM using the derivative of the VRP and possible step adjustment variables as control inputs. Employing the equation between 3D-DCM and VRP (13), the following first-order linear system is used to formulate the MPC, after the discretization.

$$\begin{aligned}\dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u},\\ \boldsymbol{\xi} &= \boldsymbol{C}\boldsymbol{x},\end{aligned} \tag{14}$$

where,

$$\begin{aligned}&\boldsymbol{x} = \begin{bmatrix}\boldsymbol{c} & \dot{\boldsymbol{c}} & \boldsymbol{p}_{vrp}\end{bmatrix}^T, \quad \boldsymbol{u} = \dot{\boldsymbol{p}}_{vrp},\\ &\boldsymbol{A} = \begin{bmatrix}\mathbf{0} & \boldsymbol{I} & \mathbf{0}\\ \frac{1}{b^2}\boldsymbol{I} & \mathbf{0} & -\frac{1}{b^2}\boldsymbol{I}\\ \mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix}\mathbf{0}\\ \mathbf{0}\\ \boldsymbol{I}\end{bmatrix},\\ &\boldsymbol{C} = \begin{bmatrix}\boldsymbol{I} & \frac{1}{b}\boldsymbol{I} & \mathbf{0}\end{bmatrix}.\end{aligned} \tag{15}$$

The formulated MPC framework is as follows:

$$\begin{aligned}\min_{\Delta\mathbf{P}_{vrp,k}\boldsymbol{f}_k}\quad & \|\boldsymbol{\Xi}_{k+1} - \boldsymbol{\Xi}_{k+1}^{ref}\|_{\mathbf{w}_\xi}^2 + \|\dot{\mathbf{P}}_{vrp,k}\|_{\mathbf{w}_p}^2\\ & + \|\boldsymbol{f}_k - \boldsymbol{f}_{k-1}\|_{\mathbf{w}_{df}}^2 + \|\mathbf{F}_k - \mathbf{F}_{k-1}\|_{\mathbf{w}_{dF}}^2\\ \text{s. t.}\quad & (VRP\ constraint)\\ & (step\ adjustment\ constraint)\\ & (intrinsic\ stability\ constraint),\end{aligned}$$

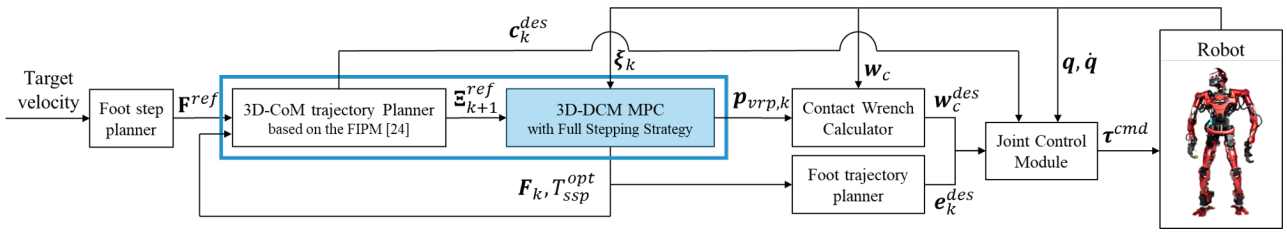


Fig. 3. Overall control framework for 3D-DCM tracking including proposed methods

TABLE I

WEIGHTING PARAMETERS FOR THE PROPOSED MPC.

$\boldsymbol{W}_{\xi}$	$\boldsymbol{W}_{p}$	$\boldsymbol{W}_{df}$	$\boldsymbol{W}_{dF}$
$[1,1,1]^{\top}$	$[0.02, 0.001, 0.01]^{\top}$	$[30,100]^{\top}$	$[100,100]^{\top}$

where $\boldsymbol{\Xi}_{k+1} \in \mathbb{R}^{3N_h}$ and $\mathbf{F}_k \in \mathbb{R}^3$ denote the predicted 3D-DCM trajectory and a step location which can be calculated from step adjustment variables, $\boldsymbol{f}_k$, respectively.

There are four terms in the cost function: 1) to track the reference 3D-DCM trajectory, 2) to regulate the derivative of VRP, 3) to regulate the difference of step adjustment variables from the previous time step $k-1$, and 4) to regulate the difference of step location from the previous time step $k-1$. As the third term regulates the derivative of the step time adjustment, the fourth term is added to regulate the step location adjustment. Weight coefficient values are provided in Table I.

The constraints comprise three components. The first constraint concerns VRP. As mentioned in Section II-B, the constraints for ZMP (2), normalized stiffness (3) are transferred to the VRP with the relation between them, (9). Similar to [15], they are formulated into a quadratic form, including the step adjustment variables. A detailed derivation is provided in the APPENDIX. The second concerns the step adjustment variables for the feasible utilization of step location and time adjustments. Lastly, intrinsic stability is considered to ensure the stability of the generated trajectory [24]–[26].

The two QCQP MPCs (3D-CoM trajectory planner, 3D-DCM MPC) are solved using Sequential Quadratic Programming (SQP) to leverage the benefits of QCQP in the calculation of the Hessian and gradient, with two iterations.

C. Full Stepping Strategy through Step Adjustment Variable

As a first step toward explaining step adjustment, a selection matrix that maps the influence of step location onto the MPC horizon is formulated as in [5], [15]. Assuming two steps within the horizon, the selection matrix for the nominal step time is defined as follows:

$$\boldsymbol{S}_{k,0} = [\underbrace{0 \cdots 0}_{p}\ \underbrace{0 \cdots 0}_{}\ \underbrace{1 \cdots 1}_{N_h-p}] \in \mathbb{R}^{N_h}. \quad (16)$$

where the first p zeroes correspond to the current step, and last $N_h - p$ ones correspond to the next step. When the current step time is reduced by one control tick, the following selection matrix can be used instead of the nominal one.

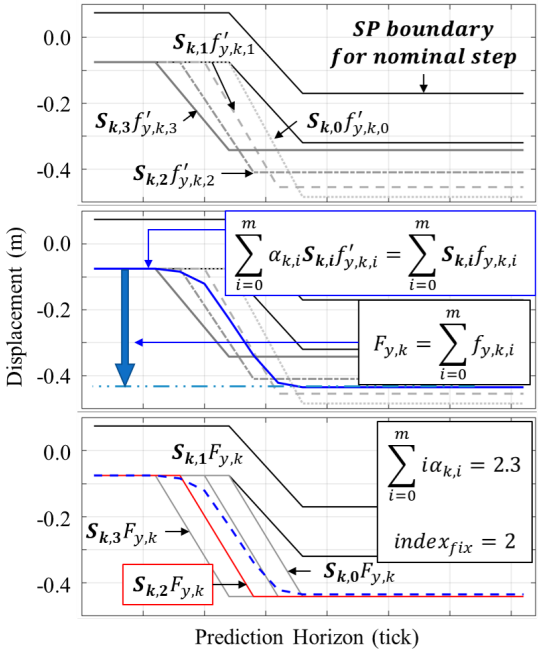


Fig. 4. The illustrations for the example of step adjustment variables and the definition of "*nearest*" when choosing the step time adjustment.

$$\boldsymbol{S}_{k,1} = [\underbrace{0 \cdots 0}_{p-1}\ \underbrace{0 \cdots 0}_{}\ \underbrace{1 \cdots 1}_{N_h-(p-1)}] \in \mathbb{R}^{N_h}. \quad (17)$$

The number of zeroes decreases to $p-1$, and the number of ones increases to $N_h - (p-1)$. More generally, a reduction of i control ticks in the current step time yields:

$$\boldsymbol{S}_{k,i} = [\underbrace{0 \cdots 0}_{p-i}\ \underbrace{1 \cdots 1}_{}\ \underbrace{1 \cdots 1}_{N_h-(p-i)}] \in \mathbb{R}^{N_h}. \quad (18)$$

Here, we emphasize that adjusting the step-time corresponds to choose to use one of the selection matrix among them. Incorporating this concept, the constraint for ZMP upper bound in (2) can be expressed as follows:

$$\mathbf{P}_{zmp} \leq \mathbf{P}_{zmp}^{max} + \sum_{i=0}^{m} \alpha_{k,i} \boldsymbol{S}_{k,i} f'_{k,i}. \quad (19)$$

where $\alpha_{k,i}$ and $f'_{k,i}$ are the binary variables and the next step location for each selection matrix. As the direct use of $\alpha_{k,i}$ and $f'_{k,i}$ introduces the trilinear term, an auxiliary variable

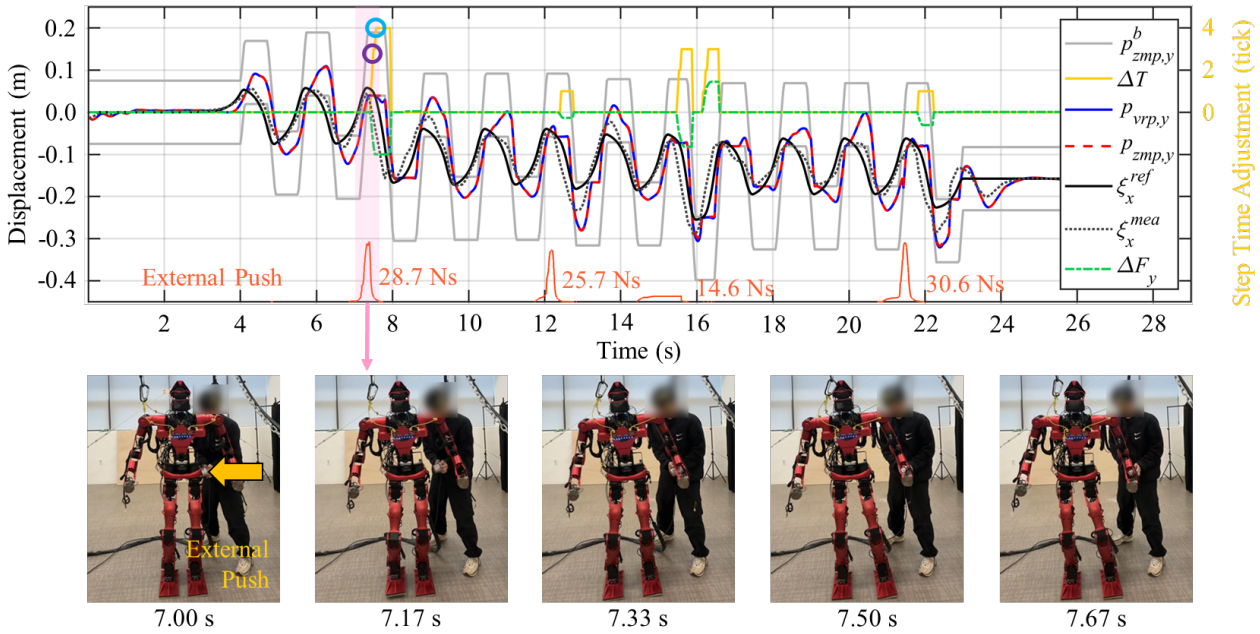


Fig. 5. Experimental Results of Proposed Method with 4 external pushes

$f_{k,i}$ is employed to modify (19) as:

$$\mathbf{P}_{zmp} \leq \mathbf{P}^{max}_{zmp} + \sum_{i=0}^{m} \boldsymbol{S}_{k,i} f_{k,i}. \tag{20}$$

The constraints between variables are as follows:

$$\begin{aligned} s.t. \sum_{i=0}^{m} \alpha_{k,i} = 1,\ \alpha_{k,i} \in \{0,1\} \\ f^{min}_k \leq f'_{k,i} \leq f^{max}_k \\ f_{k,i} = \alpha_{k,i} f'_{k,i} \\ i = 1, \ldots, m. \end{aligned} \tag{21}$$

where, f^{min}_k, f^{max}_k are the limits for the feasibility of step location. Choosing one selection matrix is enforced by the constraint that the sum of the corresponding $\alpha_{k,i}$ equals one. The step location is calculated as below.

$$F_k = \sum_{i=0}^{m} \alpha_{k,i} f'_{k,i} = \sum_{i=0}^{m} f_{k,i}. \tag{22}$$

When handling the MIP, the first procedure, based on Branch and Bound methods [27], is to solve the fully relaxed linearized problem. Following the procedure, the binary variables are relaxed as:

$$\sum_{i=0}^{m} \alpha_{k,i} = 1, \quad 0 \leq \alpha_{k,i} \leq 1. \tag{23}$$

Then, (3), (21), (20), (23), and (9) are formulated into a quadratic constraint, resulting in the QCQP.

The next step is a tree-search procedure that enumerates possible subproblems to refine the bound [27]. However, beyond this general solution strategy, it is also important to develop problem-specific simplifications that reduce computational burden by leveraging the underlying system dynamics. As shown in the third graph of Fig. 4, the strategy in this problem is to use the "*nearest*" SP constraints composed of the selection matrix to the calculated ZMP. Given that SP must encompass the computed ZMP, the definition of the "*nearest*" solution corresponds to choosing the step time adjustment with a larger extent to ensure feasibility, as below.

$$\begin{aligned} index_{fix} = round(\sum_{i=0}^{m} i\alpha_{k,i} + margin - 1) \\ T^{opt}_{step} = T^{nom}_{step} - index_{fix} \times dt_{mpc}. \end{aligned} \tag{24}$$

The illustrations for the example of step adjustment variables are shown in Fig 4. The utilized *margin* is 0.2.

IV. Results of Simulations and Experiments

A. Overview

For real robot experiments, an adult-sized humanoid robot is used. The humanoid robot has 33 DoFs: 21 in the upper body and 12 in the lower body. All actuators consist of BLDC motors and harmonic gears with a 100:1 gear ratio. The robot is 1.8 m tall and weighs 100 kg. The control computer and servo drives communicate through EtherCAT at 2kHz, and the proposed method (3D-CoM trajectory planner, 3D-DCM MPC with full stepping strategy) operates at 30 Hz in a parallel thread with a prediction horizon of $N_h = 27$. An IMU sensor is located at the pelvis, and Force/Torque (F/T) sensors are located at each ankle.

B. Push recovery

To validate the balancing performance of the proposed methods, external pushes were applied to the walking-in-place robot. The pushes were applied 4 times to the pelvis,

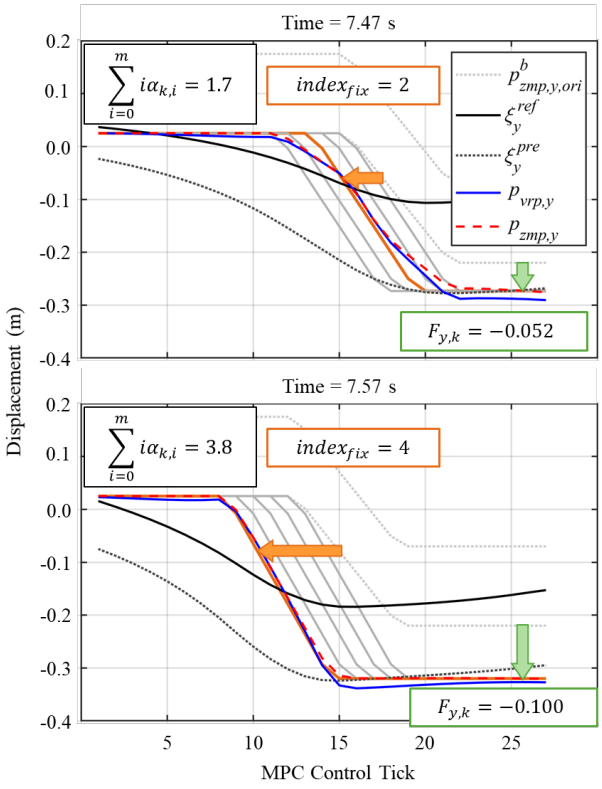


Fig. 6. MPC data for the predicted horizon at 7.47 s and 7.57 s

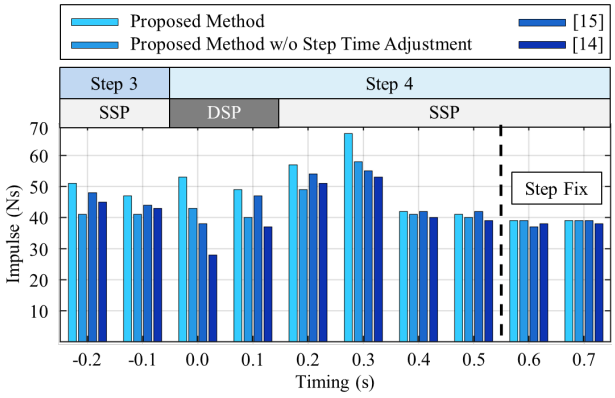


Fig. 7. Results of the comparative simulation. The maximum magnitudes of disturbance (positive y direction) that the robot can withstand are plotted.

directed from left to right (in the positive y-direction). An F/T sensor was used to measure the impact magnitude. Walking steps were planned with a step duration of 0.8 s (0.6 s Single Support Phase (SSP), 0.2 s Double Support Phase (DSP)). In Fig. 5, the y direction of ZMP boundary $p^b_{zmp,y}$, reference 3D-DCM ξ^{ref}_y, measured 3D-DCM ξ^{mea}_y, desired VRP $p^{des}_{vrp,y}$, calculated ZMP $p^{des}_{zmp,y}$ corresponding to desired VRP, step location adjustment ΔF_y, and step time adjustment ΔT are plotted as a result of the experiment.

When the first push was applied around 7 s, the desired VRP strived to maintain balance. Immediately after the corresponding ZMP reaches the boundary at 7.37 s, the pre-planned SP was insufficient to generate the VRP required to stabilize 3D-DCM, and step location and time adjustments were generated starting at 7.4 s. At 7.53 s, the step time adjustment reaches the pre-determined limit of 4 ticks, and at 7.57 s, the step location adjustment reaches the limit of -0.1 m. As the proposed MPC was operated at 30 Hz, 1 tick of the step time adjustment corresponds to 1/30 ($\approx$ 0.33) s, and 1.33 s of step time is reduced for stabilization. By implementing the calculated step adjustment and VRP, the humanoid robot maintained the balance.

To illustrate how the proposed method adjusts the step location and time, the MPC data for the predicted horizons at 7.47 s and 7.57 s (purple and sky-blue circles in Fig. 5) are shown in Fig. 6. For the MPC data, the y direction of the original ZMP boundary $p^b_{zmp,y,ori}$, reference 3D-DCM ξ^{ref}_y, predicted 3D-DCM from the measured 3D-DCM ξ^{pre}_y, predicted VRP $p_{vrp,y}$, calculated ZMP $p_{zmp,y}$ corresponding to predicted VRP for the MPC horizon are shown.

At 7.47 s, the measured 3D-DCM already went out of the original ZMP boundary, and could not be stabilized without modifying the pre-planned SP. For further utilization of VRP, the total step location adjustment of -0.052 m, $F_{y,k}$, was generated. Additionally, by enabling the step time adjustment, VRP was utilized much earlier than the pre-planned contact schedule. As the weighted average of step time adjustment variables $\alpha_{k,i}$ was 1.7, the step time adjustment was fixed to 2 ticks ($\approx$ 0.66 s) by following (24).

At 7.57 s, as 3D-DCM diverges more than at 7.47 s, it was also unable to stabilize without SP modification. The step location adjustment increased to -0.100 m, and the weighted average of step time adjustment variables increased to 3.8, which, in turn, results in 4 ticks ($\approx$ 1.33 s).

When the following pushes were applied to the robot, the proposed method calculated the desired VRP, and step location adjustments of -0.013, -0.082 (with subsequent 0.072), and -0.03 m, and step time adjustments of 1, 3 (with subsequent 3), and 1 tick to stabilize the robot.

C. Comparative Simulation

To further validate the effectiveness of step time adjustment, comparative simulations were conducted. Four controllers were utilized for the comparison: 1) the proposed method, 2) the proposed method that disabled the step time adjustment, and two 3D-DCM-based step location adjustment methods: 3) [15], 4) [14]. The external pushes were applied to the pelvis of the walking-in-place robot by varying the timing. For all controllers, walking steps were planned with a step duration of 0.8 s (0.6 s SSP, 0.2 s DSP). As a result of comparative simulation, the maximum magnitude of disturbance that the humanoid robot can withstand is shown in Fig. 7. For timing from -0.2 s, the end of step 3 SSP, to 0.3 s, the middle of step 4 SSP, the controller with step time adjustment exhibited the largest balancing capacity. For timing from 0.4 s to 0.7 s, the four controllers exhibited almost equal performance. For 0.4 to 0.5 s, the remaining time until the step fix was insufficient to generate

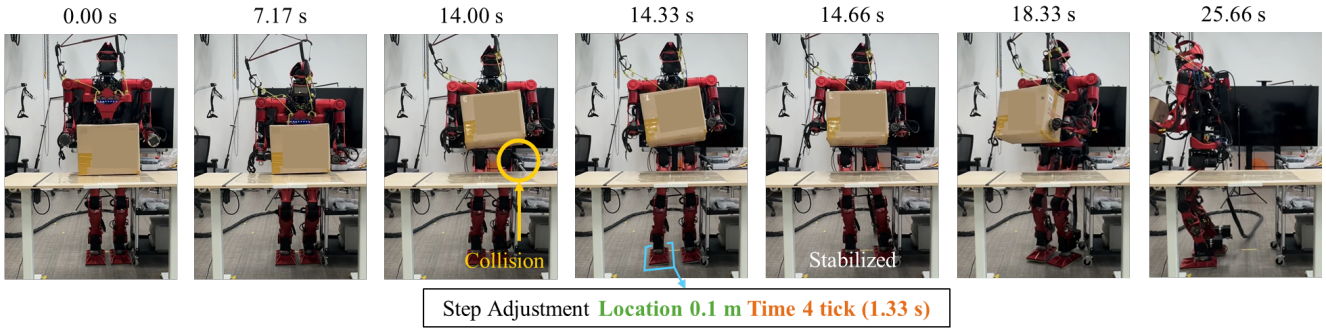


Fig. 8. Snapshots of loco-manipulation scenario experiment

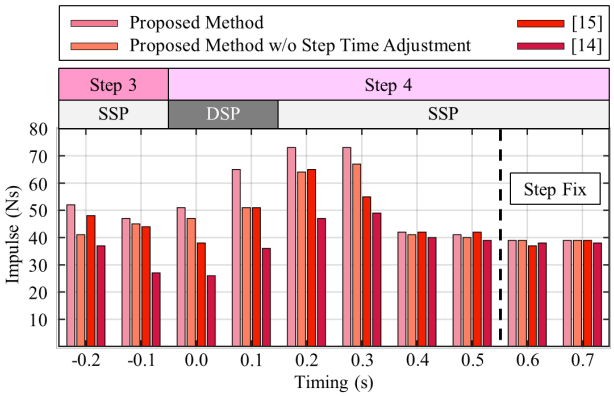


Fig. 9. Results of comparative simulations when raising CoM by 10 cm. The maximum magnitude of disturbance (positive y direction) that the robot can withstand.

the appropriate adjustment. For 0.6 s to 0.7 s, the step was fixed prior to the pushes, and no step adjustment was generated.

D. Loco-Manipulation

To verify the applicability of the proposed methods, an experiment performing a loco-manipulation scenario was conducted. In this experiment, the humanoid robot held the box on the table by lowering the CoM by 10 cm. After grasping the box, the robot restored the initial CoM height and began walking to deliver it. Assuming the worst-case scenario, an obstacle was located adjacent to the humanoid robot. The snapshots of the experiment are shown in Fig. 8. As the humanoid robot began walking during the vertical motion to restore the initial CoM height, the robot's pelvis collided with the object at 14.00 s. However, utilizing the full stepping strategy of the proposed method, a step location adjustment of 0.1 m and a step time adjustment of 4 ticks (≈ 1.33 s), the humanoid robot maintained the balance and completed the delivery scenario.

Additional comparative simulations were conducted to validate the effectiveness of step time adjustment during vertical motion. The CoM of the humanoid robot was planned to be raised by 10 cm while walking in place. As shown in the results in Fig. 9, among the four controllers in IV-C, the controller with step-time adjustment exhibited the largest balancing capacity against disturbances applied at various times, even during vertical motion.

V. CONCLUSION

In this study, a novel method that enables full stepping strategy, step location, and time adjustments, while enabling vertical motion, is proposed. Based on the FIPM and 3D-DCM dynamics, and using integer variables to represent the contact schedule in a discrete-time system, an MIP-based MPC was formulated. To alleviate the computational burden of the formulated MIP, two techniques were introduced: reducing the number of variables to improve feasibility and utilizing problem-specific simplifications. Through simulations and real-robot experiments, the balancing performance of the proposed framework is validated. However, because the resolution of the step-time adjustment is determined by the control frequency, further reducing the computational burden remains challenging. Furthermore, reinforcement learning that leverages dynamic inference from the FIPM or understanding of the step-adjustment variables will also be considered to enhance balancing capability.

APPENDIX

A. Quadratic Constraint for VRP

At first, before the derivation of quadratic constraints for VRP, the matrices to predict the future behavior of CoM and VRP is calculated based on the state equation (14).

$$\boldsymbol{C}_{x,k} = \begin{pmatrix} \boldsymbol{c}_{x,k+1} \\ \vdots \\ \boldsymbol{c}_{x,k+N_h} \end{pmatrix} = \boldsymbol{P}_{cps}\boldsymbol{x}_{x,k} + \boldsymbol{P}_{cpu}\dot{\boldsymbol{P}}_{vrp,x,k} \quad (25)$$

where

$$\boldsymbol{P}_{cps} = \begin{pmatrix} \boldsymbol{C}_{cp}\boldsymbol{A} \\ \vdots \\ \boldsymbol{C}_{cp}\boldsymbol{A}^{N_h} \end{pmatrix} \quad (26)$$

$$\boldsymbol{P}_{cpu} = \begin{pmatrix} \boldsymbol{C}_{cp}\boldsymbol{B} & \cdots & \boldsymbol{0} \\ \vdots & \ddots & \vdots \\ \boldsymbol{C}_{cp}\boldsymbol{A}^{N_h-1}\boldsymbol{B} & \cdots & \boldsymbol{C}_{cp}\boldsymbol{B} \end{pmatrix} \quad (27)$$

with

$$\boldsymbol{C}_{cp} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, \boldsymbol{x}_{x,k} = \begin{bmatrix} c_{x,k} & \dot{c}_{x,k} & p_{vrp,x,k} \end{bmatrix}^T \quad (28)$$

In a similar manner with $\boldsymbol{C}_{cv}$ and $\boldsymbol{C}_{vp}$ as below, $\boldsymbol{P}_{cvs}$, $\boldsymbol{P}_{cvu}$, $\boldsymbol{P}_{vps}$, $\boldsymbol{P}_{vpu}$ can be calculated.

$$\boldsymbol{C}_{cv} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \boldsymbol{C}_{vp} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}. \tag{29}$$

Next, selection matrices to extract variable and state from control inputs and state vectors are introduced.

$$\begin{aligned} \boldsymbol{x}_{x,k} &= \boldsymbol{S}_{sx}\boldsymbol{x}_k, & \boldsymbol{S}_{sx} &\in \mathbb{R}^{3\times 9} \\ \dot{p}_{vrp,x,k+i} &= \boldsymbol{S}_i\dot{\boldsymbol{P}}_{vrp,x,k}, & \boldsymbol{S}_i &\in \mathbb{R}^{1\times N_h} \\ \dot{\boldsymbol{P}}_{vrp,x,k} &= \boldsymbol{S}_{U_{px}}\dot{\boldsymbol{P}}_{vrp,k}, & \boldsymbol{S}_{U_{px}} &\in \mathbb{R}^{N_h\times 3N_h} \\ \dot{\boldsymbol{P}}_{vrp,k} &= \boldsymbol{S}_{U_p}\boldsymbol{U}_k, & \boldsymbol{S}_{U_p} &\in \mathbb{R}^{3N_h\times 3N_h+2n} \\ \boldsymbol{f}_{x,k} &= \boldsymbol{S}_{U_{fx}}\boldsymbol{f}_k, & \boldsymbol{S}_{U_{fx}} &\in \mathbb{R}^{n\times 2n} \\ \boldsymbol{f}_k &= \boldsymbol{S}_{U_f}\boldsymbol{U}_k, & \boldsymbol{S}_{U_f} &\in \mathbb{R}^{2n\times 3N_h+2n} \end{aligned} \tag{30}$$

where k is the time step, and i is the MPC step.

Utilizing the derivation of equation (14)-(20) in [15], following equation can be obtained.

$$\begin{aligned} &(p_{vrp,x,k+i} - c_{x,k+i})c_{z,k+i} \\ &+ (c_{z,k+i} - p_{vrp,z,k+i} + b^2 g)(c_{x,k+i} - p^{max}_{zmp,x,k+i}) \\ &\leq 0. \end{aligned} \tag{31}$$

By replacing CoM, VRP in (31) with (32), (33), and (30), quadratic constraint about control input $\boldsymbol{U}_k = [\dot{\boldsymbol{P}}_{vrp,k}, \boldsymbol{f}_k]^T$ is formulated. Noting (7), the derivative of VRP is identical to that of eCMP.

$$\begin{aligned} c_{x,k+i} &= \boldsymbol{S}_i(\boldsymbol{P}_{cps}\boldsymbol{x}_{x,k} + \boldsymbol{P}_{cpu}\dot{\boldsymbol{P}}_{vrp,x,k}) \\ &= \boldsymbol{S}_i(\boldsymbol{P}_{cps}\boldsymbol{S}_{sx}\boldsymbol{x}_k + \boldsymbol{P}_{cpu}\boldsymbol{S}_{U_{px}}\boldsymbol{S}_{U_p}\boldsymbol{U}_k), \end{aligned} \tag{32}$$

$$\begin{aligned} p_{vrp,x,k+i} &= \boldsymbol{S}_i(\boldsymbol{P}_{vps}\boldsymbol{x}_{x,k} + \boldsymbol{P}_{vpu}\dot{\boldsymbol{P}}_{vrp,x,k}) \\ &= \boldsymbol{S}_i(\boldsymbol{P}_{vps}\boldsymbol{S}_{sx}\boldsymbol{x}_k + \boldsymbol{P}_{vpu}\boldsymbol{S}_{U_{px}}\boldsymbol{S}_{U_p}\boldsymbol{U}_k). \end{aligned} \tag{33}$$

For the 3D-DCM MPC, the boundary for ZMP is calculated with step adjustment variable, still formulating the quadratic constraints about control input $\boldsymbol{U}_k = [\dot{\boldsymbol{P}}_{vrp,k}, \boldsymbol{f}_k]^T$ is formulated. Corresponding equations can be derived for lower boundary and y direction.

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